Institute for
Data Science Foundations
E-21

Prof. Dr. Nihat Ay

Lecture 06

Statistical Estimation

Recall that a statistical model M is a paramatrised family of probability distributions:

$$\Theta \ni \theta \mapsto p_\theta \in \mathcal{P}_+ \,. \tag{1}$$

Examples are Bernoulli model (e.g. coin toss) and a normal model (e.g. height distribution of humans).

Suppose we are given a sample x from the distribution p_{θ^*}, then we are interested in estimating the value of θ^*. The topic of today will be to investigate how the Fisher Information Matrix is related to this estimation problem.

Fisher Information Matrix

The Fisher Information Matrix (FIM) is an inner product on the parameter space. We can use it to compute the length $L(v)$ of a tangent vector $v \in T_\theta\Theta$ as follows:

$$L(v) = ||v||_\theta = \sqrt{v^T G(\theta) v}. \tag{2}$$

The lenght of a unit vector $e_i = \frac{\partial}{\partial \theta_i} = (0, ..., 0, \underbrace{1}_{i\text{th entry}}, 0, ..., 0)$ is therefore given by

$$L(e_i) = \sqrt{G_{ii}(\theta)} \tag{3}$$

We can compute the length of a curve in the parameter space by

$$L(\gamma) = \int_0^1 ||\dot{\gamma}(t)||_{\gamma(t)} dt. \tag{4}$$

The Fisher distance and FIM are related as follows:

$$\left.\frac{d}{dt}\right|_{t=0} d^F(p_\theta, p_{\theta+tv}) = \sqrt{v^T G(\theta) v} \tag{5}$$

Coin toss

Consider the Bernoulli (coin toss) model where $I = \{0, 1\}$ and

$$p_\theta = (1-\theta)\delta^0 + \theta\delta^1 \tag{6}$$

$$\tag{7}$$

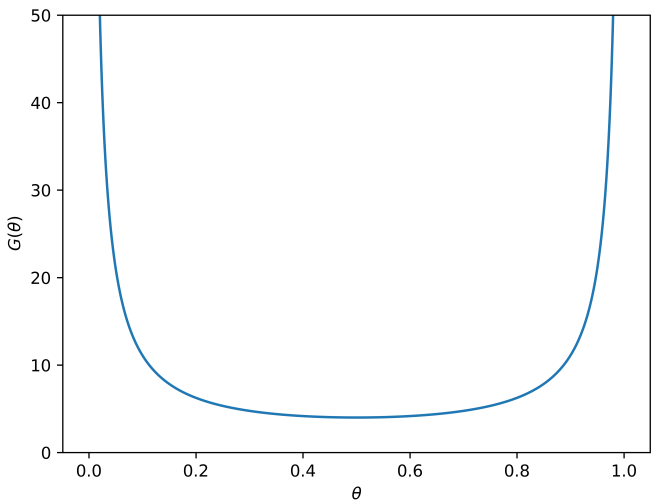

Figure 1: FIM for the Bernoulli model

The Fisher Information Matrix for this model is given by:

$$G(\theta) = \frac{1}{\theta(1-\theta)} \tag{8}$$

See Figure 1

The Fisher distance tells us how different two probability distributions are, i.e. how easily we can distinguish between them on the basis of samples. (more on this later).

Normal distributions

Consider the model:

$$(\theta_1, \theta_2) = (\mu, \sigma) \mapsto p(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{9}$$

In Lecture 3 we computed

$$G_{11}(\theta) = \frac{1}{\sigma^2} \tag{10}$$

The length of the curve $\gamma(t) = ((1-t)\mu + t\tilde{\mu}, \sigma)$ with $\tilde{\mu} > \mu$ is therefore given by $\frac{\tilde{\mu}-\mu}{\sigma}$. Note that this curve becomes shorter when σ is increased, which corresponds to our intuition that normal distributions with large variance are more difficult to distinguish from one another.

Multiple samples

Back to our estimation problem. Normally we sample not a single datum x but multiple independent data $x^N = (x^{(1)}, ..., x^{(N)})$ from a distribution and try to estimate θ^* from this.

Our (product) model is now as follows:

$$\theta \mapsto p(x^N; \theta) \tag{11}$$

$$p(x^N; \theta) = \prod_i^N p(x^{(i)}; \theta) \tag{12}$$

Example: two coin tosses

Our sample space is now given by

$$I^2 = \{(0,0), (0,1), (1,0), (1,1)\} \tag{13}$$

We have

$$p_\theta(x_1, x_2) = \begin{cases} (1-\theta)^2 & \text{if } x_1 = x_2 = 0 \\ (1-\theta)\theta & \text{if } x_1 = 0, x_2 = 1 \text{ or } x_1 = 1, x_2 = 0 \\ \theta^2 & \text{if } x_1 = x_2 = 1 \end{cases} \tag{14}$$

Note that this is a one-dimensional submanifold of the three-dimensional probability simplex $\mathcal{P}_+(I^2)$. Furthermore this is also a submanifold of the independence model discussed previously.

The Fisher matrix for this product model is given by

$$G_{ij}^N(\theta) = -\mathbb{E}_{p_\theta(x^N)}\left(\frac{\partial^2}{\partial\theta_i\partial\theta_j} \ln p(x^N; \theta)\right) \tag{15}$$

Use,

$$\ln p(x^N; \theta) = \sum_i^N \ln p(x^{(i)}; \theta) \tag{16}$$

$$\tag{17}$$

$$\begin{aligned} G_{ij}^N(\theta) &= -\mathbb{E}_{p_\theta(x^N)}\left(\frac{\partial^2}{\partial\theta_i\partial\theta_j} \ln p(x^N; \theta)\right) && (18) \\ &= -\mathbb{E}_{\prod_i^N p_\theta(x^{(i)})}\left(\frac{\partial^2}{\partial\theta_i\partial\theta_j} \sum_i^N \ln p(x^{(i)}; \theta)\right) && (19) \\ &= -\sum_i^N \mathbb{E}_{p_\theta(x^{(i)})}\left(\frac{\partial^2}{\partial\theta_i\partial\theta_j} \ln p(x^{(i)}; \theta)\right) && (20) \\ &= NG_{ij}(\theta) && (21) \end{aligned}$$

From now on we sometimes suppress the N in the notation.

Sufficient Statistics

For a finite sample space I from which we draw N samples, a statistic is a map

$$f : I^N \to I' \tag{22}$$

Note that this map induces a map

$$f_* : \mathcal{P}(I^N) \to \mathcal{P}(I') \tag{23}$$

$$f_*(p^N(\cdot))(y) = p^f_\theta(y) = \sum_{x^N \in f^{-1}(y)} p(x^N). \tag{24}$$

And this also gives us a new model

$$\theta \mapsto p^f(y) \in \mathcal{P}(I') \tag{25}$$

The corresponding Fisher information matrix is written G^f.

Examples: (coin tosses)

$$f(x_1, x_2) = x_1 + x_2 \qquad \in \{0, 1, 2\} \tag{26}$$

$$f(x_1, x_2) = x_1 \qquad \in \{0, 1\} \tag{27}$$

$$f(x_1, x_2) = (x_1 + x_2) \mod 2 \qquad \in \{0, 1\} \tag{28}$$

We are now interested in the question: "When do we lose information about θ^* when we observe $y = f(x^N)$ instead of x^N directly?".

Think about the three examples

A statistic is called *sufficient* when the distribution of x^N given $f(x^N)$ is independent of θ. **work out for examples**

Theorem. *For a general statistic f we have:*

$$G^N(\theta) \geq G^f(\theta). \tag{29}$$

A statistic is sufficient if and only if:

$$G^N(\theta) = G^f(\theta). \tag{30}$$

Remark. For matrices M, N we write $M \geq N$ when the $M - N$ is positive semi-definite.

Therefore the Fisher information matrix can be used to quantify how much information is lost by observing $f(x^N)$ instead of x^N. This comes back to the statement that the FIM expresses how easily two distribution p_θ and $p_{\tilde{\theta}}$ can be distinguished on the basis of a sample.

Estimators

An *estimator* $\hat{\theta}$ is defined as follows:

$$\hat{\theta} : \Omega^N \to \Theta \tag{31}$$

Example: estimate average height

How to determine what is a good estimator?

An estimator $\hat{\theta}$ is *unbiased* if

$$\mathbb{E}_{p_{\theta^*}(x^N)}\left[\hat{\theta}(x^N)\right] = \theta^* \tag{32}$$

For an unbiased estimator we have that the mean squared error can be expressed as the covariance matrix as follows:

$$\mathbb{V}^{ij}_{\theta^*}[\hat{\theta}] = \mathbb{E}_{p_{\theta^*}(x^N)}\left[(\hat{\theta}_i(x^N) - \theta^*_i)(\hat{\theta}_j(x^N) - \theta^*_j)\right] \tag{33}$$

We have the following result involving the FIM.

Theorem. *The variance matrix $\mathbb{V}_{\theta^*}[\hat{\theta}]$ of an unbiased estimator $\hat{\theta}$ satisfies the following condition:*

$$\mathbb{V}_{\theta^*}[\hat{\theta}] \geq G^{-1}(\theta). \tag{34}$$

An estimator is called *efficient* if it satisfies $\mathbb{V}_{\theta^*}[\hat{\theta}] = G^{-1}(\theta)$.

In the words of Amari:

"The matrix $G^{-1}(\theta)$ represent the degree to which an optimal estimator fluctuates around the true value of the parameter θ^*. In other words, a smaller $G^{-1}(\theta)$ indicates a more accurate estimator. To be able to accurately estimate the parameter θ^* means that as the value of θ is changed, the "character" (i.e. p_θ) of the data changes dramatically. The FIM may be considered to be a geometric expression of the size of this change."